ACSC/STAT 418, STAT 818: Time Series Analysis and Forecasting Computer Lab
Fall 2026

Territorial Acknowledgement

The University of Regina is situated on the territories of the nêhiyawak, Anihšināpēk, Dakota, Lakota, Nakoda, and the homeland of the Métis/Michif Nation. The Regina campus is on Treaty 4 lands, and Saskatoon classes are on Treaty 6 lands.

Instructor and Course Information

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Office Hours:

Wednesday 2:30 - 3:30 p.m. or by appointment
If you are planning to attend during office hours at this time, please send me a text message. There maybe some pop-up administrative meetings on Wednesdays.

Class Schedule: Tuesday and Thursday 10:00 - 11:15 a.m. in CL 305 and remotely on Zoom

https://uregina-ca.zoom.us/j/94810647841?pwd=OU5OQWNFTm0rUkVBdFJJa0VWbGZUUT09

Note: All times listed refer to times in Regina, Saskatchewan. Saskatchewan does **not** have daylight savings time. If you are in a location where the clocks change be aware of your local time of day change for the class.

Lab Information:

Lab Instructor: Sarah Carnochan Naqvi
Email: mailto:Sarah.Carnochan.Naqvi@uregina.ca or through URCourses

There will be 8 weekly labs offered asynchronously with a corresponding lab assignment. Students will need to have access to a computer with R and RStudio installed. For any difficulties, contact the Lab Instructor. All lab resources, lab office hours, zoom links, etc. will be posted in a dedicated lab URCourse.

Course Description

This course aims to introduce various statistical models for time series and cover the main methods for analysis and forecasting. Topics include: Deterministic time series: Trends and Seasonality; Random walk models; Stationary time series: White noise processes, Autoregressive (AR), Moving Average (MA), Autoregressive Moving Average (ARMA) models; Estimation, Diagnosis and Forecasting with various time series models; computer programming for Time Series Analysis.

Books

Students should be able to access the following books and references from the university library. If you encounter any difficulties, please contact the course instructor.

Textbook: Cowpertwait, P.S.P. and Metcalfe, A.V., *Introductory Time Series with R*, Springer.

Optional Textbooks:

1. Brockwell, P.J. and Davis, R.A., *Introduction to Time series and Forecasting*, Second Edition, Springer, Ch. 1 – 6.
2. Pindyck, R.S. and Rubinfeld, D.L., *Econometric Models and Economic Forecasts*, Fourth Edition, Irwin McGraw-Hill, Boston, Part 4 Time-Series Models (Ch.15 – 19).

Supplementary study material: McLeod, A. I., Yu, H. and Mahdi, E., *Time Series Analysis with R, The Handbook of Statistics*, Vol. 30, Elsevier.

Grading

- Assignments 10%, Term project 10%, Computer Lab 15%, Midterm 20%, Final 45%.
- There will be five assignments and one midterm TBA. The final exam is on December 10, 2026, 9:00 a.m. Location for on-campus students is CL 305. Location for remote students is to be determined.
- The midterm exam will be held at the normal class time and location for on-campus students. The midterm exam for remote students will be during the normal class time.
- There will be no makeup midterm under any circumstances. If you will miss the midterm, the corresponding weight will be transferred to the final exam.
- Term Project: Students will work in groups of two (or individually) to carry out one term project involving applications of the time series models and methods covered in this course. The project report is due on the 2nd of December. A more detailed guideline will be handed out later.

Other Information

- Course attendance: Students are responsible for all material covered in the lectures.
- Any notifications and course materials such as lecture notes, assignments and solutions, will be posted on the Course URCourse. All lab notifications, lab materials, lab assignments and solutions will posted on the Lab URCourse.

Academic Integrity

Academic integrity requires students be honest. Assignments and exams are to help students learn; grades show how fully this goal is attained. Thus, all work and grades should result from a student's own understanding and effort.

Plagiarism is a part of Academic Misconduct. Students must write their assignments and project report (as a team or individual) in their own words. Whenever students take an idea or a passage from another author, they must acknowledge their debt both by using quotation marks and by proper referencing such as footnotes or citations. Copying from ChatGPT or another AI is strictly prohibited.

Instances of academic misconduct will be reported to the Associate Dean Academic for investigation. Full details are provided in the appropriate calendar.

Undergraduate Academic Calendar:

https://www.uregina.ca/registrar/academic-calendars-and-schedule/undergraduate-calendar.html.

Graduate Academic Calendar:

https://www.uregina.ca/graduate-studies-research/graduate-calendar/uni-policies-procedures.html#aca2.

Students are encouraged to understand your obligations as a student, as well as your rights.

Accommodations

The Centre for Student Accessibility upholds the University's commitment to a diverse and inclusive learning environment by providing services and supports for students based on disability, religion, family status, and gender identity. Students who require these services are encouraged to contact the Centre for Student Accessibility to discuss the possibility of academic accommodations and other supports as early as possible. For further information, please email:

accessibility@uregina.ca.